

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

# Create dataset
data = {
    'Year': list(range(2010, 2021)),
    'Job Postings': [150, 300, 450, 600, 800, 1200,
                     1600, 2100, 2700, 3400, 4200]
}

# Create DataFrame
df = pd.DataFrame(data)

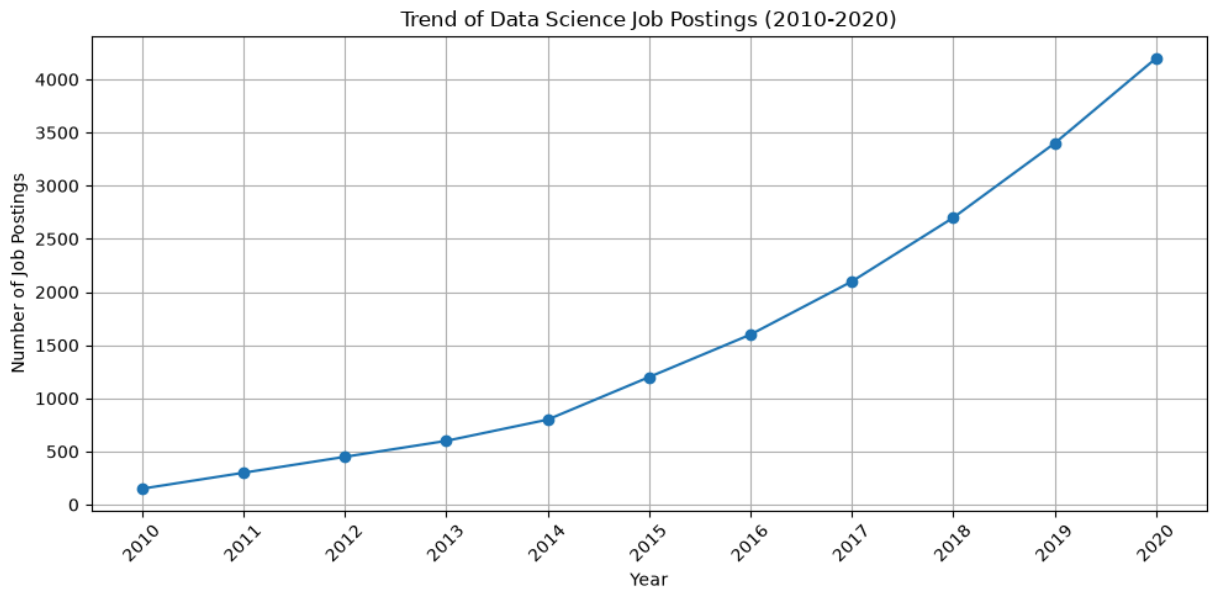
# Display dataset
print("Data Science Job Postings:")
print(df)

# Plot the trend
plt.figure(figsize=(10, 5))
plt.plot(df['Year'], df['Job Postings'],
         marker='o', linestyle='-')

plt.title('Trend of Data Science Job Postings (2010-2020)')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.xticks(df['Year'], rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```

Data Science Job Postings:

	Year	Job Postings
0	2010	150
1	2011	300
2	2012	450
3	2013	600
4	2014	800
5	2015	1200
6	2016	1600
7	2017	2100
8	2018	2700
9	2019	3400
10	2020	4200



```
In [2]: import matplotlib.pyplot as plt

roles = ['Data Analyst', 'Data Engineer', 'Data Scientist',
         'ML Engineer', 'Business Analyst']

counts = [300, 500, 450, 200, 150]

plt.figure(figsize=(9, 5))
plt.bar(roles, counts)

plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



```
In [3]: import pandas as pd
import json

# Structured data example
structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})

print("Structured Data:\n")
print(structured_data)

# Unstructured data example
unstructured_data = (
    "This is an example of unstructured data. "
    "It can be a piece of text, an image, or a video file."
)

print("\nUnstructured Data:\n")
print(unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {
    'ID': 1,
    'Name': 'Alice',
    'Attributes': {
        'Height': 165,
        'Weight': 68
    }
}

print("\nSemi-structured Data:\n")
print(json.dumps(semi_structured_data, indent=4))
```

Structured Data:

	ID	Name	Age
0	1	Alice	25
1	2	Bob	30
2	3	Charlie	35

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:

```
{
  "ID": 1,
  "Name": "Alice",
  "Attributes": {
    "Height": 165,
    "Weight": 68
  }
}
```

```
In [4]: from cryptography.fernet import Fernet

# Generate encryption key
key = Fernet.generate_key()
cipher_suite = Fernet(key)

# Original sensitive data
plain_text = b"Rajalakshmi Engineering College."

# Encrypt the data
cipher_text = cipher_suite.encrypt(plain_text)

# Decrypt the data
decrypted_text = cipher_suite.decrypt(cipher_text)

# Display the results
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

Original Data: b'Rajalakshmi Engineering College.'

Encrypted Data: b'gAAAAABqyILuP1RrQQR_4B0uIR1z5979VK3-379Pu08yiW_TyCeVe4tIbGZw2PjdOEZKUDPqkER7bXfuj4aps2iXL7-tWU7nD4USgg6IbMXY28t8x97M8zBF0aXVPtrJqINUVjhl2ER6'

Decrypted Data: b'Rajalakshmi Engineering College.'

In []: